

The Statistics.

Normal(μ, σ^2) L	Normal distribution (Gaussian distribution)	$f(x; \mu, \sigma^2) = \frac{1}{\sqrt{2\pi\sigma^2}} * \exp(-((x - \mu)^2 / (2\sigma^2)))$	$\mu = (1 / N) * \sum x_i$ (from $i=1$ to N); $\sigma^2 = (1 / N) * \sum (x_i - \mu)^2$ (from $i=1$ to N); $\sigma = \sqrt{\sigma^2}$	Models data clustering around a mean, with probabilities decreasing symmetrically as values deviate. Essential for statistical inference, hypothesis testing, and machine learning. Driven by the Central Limit Theorem (CLT).
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Standard normal	Standard normal distribution	$z = (x - \mu) / \sigma$; $\Phi(z) = P(Z \leq z)$; $f(z) = (1 / \sqrt{2\pi}) * e^{(-z^2 / 2)}$	$\mu = 0$; $\sigma^2 = 1$; $\sigma = 1$. Use z-tables or software to find probabilities.	A universal reference for measuring deviations from the mean. Allows transformation of any normal distribution to a common scale via standardization (Z-score). Enables probability estimation and comparisons across datasets.
Uniform(a, b) R,V	Uniform distribution	$f(x; a, b) = 1 / (b - a)$	$a = \min$; $b = \max$	All outcomes are equally likely within a range.

Standard uniform V	Standard uniform distribution	$f(x) = 1, \text{ for } 0 \leq x \leq 1$	No parameters	A simple uniform distribution between 0 and 1, used in probability modeling.
Exponential (a) F,M,S,V	Exponential distribution	$f(x; a) = a * \exp(-a x)$	a = rate parameter	Used in modeling time between events in a Poisson process.
Standard Wald(λ) S	Standard Wald distribution	$f(x; \lambda) = (\lambda / (2\pi x^3))^{(1/2)} * \exp(-\lambda (x - 1)^2 / (2 x))$	λ = shape parameter	A special case of the inverse Gaussian distribution with $\mu = 1$.
Rayleigh(a) M,S,V	Rayleigh distribution	$f(x; a) = (x / a^2) * \exp(-x^2 / (2 a^2))$	a = scale parameter	Used in modeling wave heights and wind speeds.
Cauchy(a, α) C,I,S,V	Cauchy distribution	$f(x; a, \alpha) = (1 / (\pi \alpha)) * (1 / (1 + ((x - a) / \alpha)^2))$	a = location parameter; α = scale parameter	Used in modeling data with heavy tails and undefined variance.

Standard Cauchy I, S, V	Standard Cauchy distribution	$f(x) = (1 / \pi) * (1 / (1 + x^2))$	No parameters	A special case of the Cauchy distribution with $a = 0$ and $\alpha = 1$.
Laplace(a_1, a_2) V	Laplace distribution	$f(x; a_1, a_2) = (1 / (2 a_2)) * \exp(- x - a_1 / a_2)$	a_1 = location parameter; a_2 = scale parameter	Used
Logistic(λ, κ) S, V	Logistic distribution	$f(x; \lambda, \kappa) = \exp(-(x - \lambda)/\kappa) / (\kappa(1 + \exp(-(x - \lambda)/\kappa))^2)$	λ = location parameter; κ = scale parameter	Used in machine learning, logistic regression, and growth modeling.
Standard power (β) V, X	Standard power distribution	$f(x; \beta) = \beta x^{(\beta-1)}$	β = shape parameter	Models power-law behavior in physics and economics.
Power (a, β) S, V, X_aution. Hyperbolic-secant	Power distribution	$f(x; a, \beta) = \beta a^\beta x^{(\beta-1)}$	a = scale; β = shape	Used in modeling tail-heavy distributions.
Arcsin V	Arcsin distribution	$f(x) = 1 / (\pi \sqrt{x(1-x)})$	No parameters	Used in modeling probabilities over bounded intervals.

Hyperbolic -secant V	Hyperbolic -secant distribution	$f(x) = (1 / 2) * \operatorname{sech}(\pi x / 2)$	No parameters	A symmetric distribution with heavy tails.
Bernoulli(p) M P X	Bernoulli distribution	$f(x; p) = p^x * (1 - p)^{(1 - x)}$	p = success probability	Models a single trial with two possible outcomes (success/ failure).
Binomial(n, p) C_p	Binomial distribution	$f(x; n, p) = C(n, x) * p^x * (1 - p)^{(n - x)}$	n = number of trials; p = success probability	Models the number of successes in a fixed number of independen t trials.
Poisson(μ) C	Poisson distribution	$f(x; \mu) = (e^{(-\mu)} * \mu^x) / x!$	μ = mean number of events	Models the number of events occurring in a fixed interval of time or space.
Geometric(p) F,M,V	Geometric distribution	$f(x; p) = p * (1 - p)^{(x - 1)}$	p = success probability	Models the number of trials until the first success.

Discrete uniform(a,b) R, V	Discrete uniform distribution	$f(x; a, b) = 1 / (b - a + 1)$	a = minimum value; b = maximum value	All outcomes are equally likely within a finite range.
Rectangular(n) V	Rectangular distribution	$f(x; n) = 1 / n$	n = range of the distribution	A uniform distribution over a finite set of values.
Logarithmic(c)	Logarithmic distribution	$f(x; c) = (-1 / \log(1 - c)) * (c^x / x)$	c = parameter (0 < c < 1)	Used in modeling rare events.
Benford V	Benford distribution	$f(x) = \log_{10}(1 + 1/x)$	No parameters	Models the frequency of first digits in naturally occurring datasets.
Chi(n)	Chi distribution	$f(x; n) = (2^{(1-n/2)} * x^{(n-1)} * e^{(-x^2/2)}) / \Gamma(n/2)$	n = degrees of freedom	Used in hypothesis testing and confidence interval estimation for standard deviation in normal distributions.

Chi-square(n) C	Chi-square distribution	$f(x; n) = \frac{(x^{n/2-1} * e^{-x/2})}{(2^{n/2} * \Gamma(n/2))}$	n = degrees of freedom	Used in goodness-of-fit tests, tests of independence, and variance estimation.
Noncentral chi-square(n, δ) C	Noncentral chi-square distribution	$f(x; n, \delta) = \sum [\frac{(e^{-x/2} * (\delta/2)^k / k!)}{(x^{n/2+k-1} * e^{-x/2})} / (2^{n/2+k} * \Gamma(n/2 + k))]$	n = degrees of freedom; δ = noncentrality parameter	Used in power analysis and noncentral hypothesis testing.
t(n)	t-distribution	$f(x; n) = \frac{(\Gamma((n+1)/2) / (\sqrt{n\pi}) \Gamma(n/2)) * (1 + x^2/n)^{-(n+1)/2}}$	n = degrees of freedom	Used in small-sample hypothesis testing and confidence intervals.
Noncentral t(n, δ)	Noncentral t-distribution	$f(x; n, \delta) = \frac{(n^{n/2} * e^{-\delta^2/2})}{(\Gamma(n/2) * 2^{n/2-1})} * \sum [\frac{(\delta^k / k!)}{(x^{n+k-1} / (1 + x^2)^{(n+k+1)/2})}]$	n = degrees of freedom; δ = noncentrality parameter	Used in power analysis and noncentral hypothesis testing.

Doubly noncentral $t(n, \delta, \gamma)$	Doubly noncentral t-distribution	$f(x; n, \delta, \gamma) = \frac{(n^{n/2} \cdot e^{-(\delta^2 + \gamma^2)/2})}{(\Gamma(n/2) \cdot 2^{n/2 - 1})} \cdot \sum_{k=0}^{\infty} \left[\frac{(\delta^2)^k}{k!} \cdot \frac{\gamma^{n-k}}{(n-k)!} \cdot \frac{x^{n-k-1}}{(1+x^2)^{(n-k+1)/2}} \right]$	n = degrees of freedom; δ , γ = noncentrality parameters	Used in advanced power analysis and noncentral hypothesis testing.
Erlang(a,n) S	Erlang distribution	$f(x; a, n) = \frac{(a^n \cdot x^{n-1} \cdot e^{-a x})}{(n-1)!}$	a = rate parameter; n = shape parameter	A special case of the gamma distribution used in queuing theory and telecommunications.
Gamma(a,β)) C_a, S	Gamma distribution	$f(x; a, \beta) = \frac{(\beta^a \cdot x^{a-1} \cdot e^{-\beta x})}{\Gamma(a)}$	a = shape parameter; β = scale parameter	Models waiting times, insurance claims, and rainfall amounts.

Inverted gamma(α, β)	Inverted gamma distribution	$f(x; \alpha, \beta) = (\beta^\alpha / \Gamma(\alpha)) * x^{(-\alpha-1)} * e^{(-\beta/x)}$	α = shape parameter; β = scale parameter	Used in Bayesian inference as a conjugate prior for the variance of a normal distribution .
Log gamma(α, β)	Log-gamma distribution	$f(x; \alpha, \beta) = (\beta^\alpha / \Gamma(\alpha)) * (\log x)^{(\alpha-1)} * x^{(-\beta-1)}$	α = shape parameter; β = rate parameter	Used in modeling positively skewed data.
Beta(β, γ)	Beta distribution	$f(x; \beta, \gamma) = (x^{(\beta-1)} (1-x)^{(\gamma-1)}) / B(\beta, \gamma)$	β, γ = shape parameters	Used in Bayesian statistics and proportions modeling.
Noncentral beta(β, γ, δ)	Noncentral beta distribution	$f(x; \beta, \gamma, \delta) = \sum [(e^{(-\delta/2)} * (\delta/2)^k / k!) * (x^{(\beta+k-1)} * (1-x)^{(\gamma-1)}) / B(\beta+k, \gamma)]$	β, γ = shape parameters ; δ = noncentrality parameter	Used in power analysis and noncentral hypothesis testing.

Inverse Gaussian(λ, μ) L	Inverse Gaussian distribution	$f(x; \lambda, \mu) = (\lambda / (2\pi x^3))^{1/2} * \exp(-(\lambda (x - \mu)^2) / (2 \mu^2 x))$	λ = shape parameter; μ = mean	Used in modeling positively skewed data, such as lifetimes or financial data.
Weibull(a, β) M- β , S, V	Weibull distribution	$f(x; a, \beta) = (\beta / a) * (x / a)^{\beta - 1} * \exp(-(x / a)^\beta)$	a = scale parameter; β = shape parameter	Used in reliability analysis and survival modeling.
Extreme value(a, β) V- i , M- β	Extreme value distribution	$f(x; a, \beta) = (1 / \beta) * \exp(-(x - a)/\beta - \exp(-(x - a)/\beta))$	a = location parameter; β = scale parameter	Used in modeling extreme events, such as floods or earthquakes.
Generalized Pareto (δ, κ, γ)	Generalized Pareto distribution	$f(x; \delta, \kappa, \gamma) = (1 / \delta) * (1 + \kappa (x - \gamma) / \delta)^{-1 - 1/\kappa}$	δ = scale parameter; κ = shape parameter; γ = location parameter	Used in modeling extreme value data and tail behavior.

Pareto(λ, κ)	Pareto distribution	$f(x; \lambda, \kappa) = (\kappa \lambda^\kappa) / x^{(\kappa+1)}$	λ = scale parameter; κ = shape parameter	Models wealth distribution , city populations , and file sizes in computer networks.
Log logistic(λ, κ) I, S, V	Log-logistic distribution	$f(x; \lambda, \kappa) = (\kappa / \lambda) * (x / \lambda)^{(\kappa - 1)} / (1 + (x / \lambda)^\kappa)^2$	λ = scale parameter; κ = shape parameter	Used in survival analysis and reliability modeling.
Lomax(λ, κ) V	Lomax distribution	$f(x; \lambda, \kappa) = (\kappa / \lambda) * (1 + x / \lambda)^{-(\kappa+1)}$	λ = scale parameter; κ = shape parameter	A variant of the Pareto distribution used in business failure modeling.
Inverted beta(β, γ)	Inverted beta distribution	$f(x; \beta, \gamma) = x^{(-\beta-1)} / B(\beta, \gamma)$	β, γ = shape parameters	Used in Bayesian inference and financial modeling.
Error(a,b,c) S	Error distribution	$f(x; a, b, c) = (c / (2b)) * \exp(- x - a / b)$	a = location; b = scale; c = shape parameter	Used in modeling symmetric noise in data.

Standard triangular V	Standard triangular distribution	$f(x; a, b, m) = 2(x-a) / ((b-a)(m-a)), a \leq x \leq m; 2(b-x) / ((b-a)(b-m)), m \leq x \leq b$	$a = \min; b = \max; m = \text{mode}$	Used in project management and simulations.
Triangular(a,b,m) V	Triangular distribution	$f(x; a, b, m) = 2(x-a) / ((b-a)(m-a)), a \leq x \leq m; 2(b-x) / ((b-a)(b-m)), m \leq x \leq b$	$a = \min; b = \max; m = \text{mode}$	Used in project planning and risk analysis.
TSP(a,b,m,n) V	TSP distribution	$f(x; a, b, m, n) = (b-a) / ((x-a)(b-x))^m * n$	$a, b = \text{bounds}; m, n = \text{shape parameters}$	Used in reliability modeling and life expectancy calculations.
Arctangent(λ, ϕ) S, V	Arctangent distribution	$f(x; \lambda, \phi) = (1 / \pi) * (\lambda / (\lambda^2 + (x - \phi)^2))$	$\lambda = \text{scale parameter}; \phi = \text{location parameter}$	Used in modeling data with heavy tails.
von Mises(κ, μ) S	von Mises distribution	$f(x; \kappa, \mu) = (e^{(\kappa \cos(x - \mu))}) / (2\pi I_0(\kappa))$	$\kappa = \text{concentration}; \mu = \text{mean}$	Used in directional statistics for circular data.

Generalized gamma(α, β, γ)	Generalized gamma distribution	$f(x; \alpha, \beta, \gamma) = (\gamma / (\beta^\alpha \Gamma(\alpha))) * x^{(\alpha\gamma-1)} * e^{-(x/\beta)^\gamma}$	α = shape parameter; β = scale parameter; γ = shape parameter	A flexible distribution for modeling various types of data.
Log normal(α, β) P	Log-normal distribution	$f(x; \alpha, \beta) = (1 / (x \beta \sqrt{2\pi})) * \exp(-((\log x - \alpha)^2 / (2\beta^2)))$	α = mean of $\log(X)$; β = standard deviation of $\log(X)$	Used for modeling data that are positively skewed, such as income or stock prices.
Exponential power (λ, κ) V	Exponential power distribution	$f(x; \lambda, \kappa) = (\kappa / (2\lambda \Gamma(1/\kappa))) e^{-(x /\lambda)^\kappa}$	λ = scale; κ = shape	Generalization of normal and Laplace distributions for flexible modeling.
Minimax(β, γ)	Minimax distribution	$f(x; \beta, \gamma) = (\gamma / (\beta^\gamma)) x^{(\gamma-1)}$	β, γ = shape parameters	Used in decision theory and risk assessment.

IDB(δ, κ, γ)	IDB distribution	$f(x; \delta, \kappa, \gamma) = (\delta \kappa^\gamma x^{\gamma-1} \exp(-\kappa x)) / \Gamma(\gamma)$	δ, κ, γ = shape and scale parameters	Used in reliability analysis and survival modeling.
Muth(κ)	Muth distribution	$f(x; \kappa) = \kappa * \exp(-\kappa x)$	κ = rate parameter	Used in reliability analysis and survival modeling.
Negative binomial(n, p)				
Pascal(n, p) C_p	Pascal distribution	$f(x; n, p) = C(x-1, n-1) * p^n * (1-p)^{(x-n)}$	n = number of successes; p = success probability	Models the number of trials until a specified number of successes.
Polya(n, p, β)	Polya distribution	$f(x; n, p, \beta) = (\Gamma(n+x) / (\Gamma(x+1) \Gamma(n))) * p^x * (1-p)^n$	n = number of trials; p = success probability; β = overdispersion parameter	Used in modeling overdispersed count data.
Zipf(a, n)	Zipf distribution	$f(x; a, n) = (1 / x^a) / H(n, a)$	a = exponent; n = number of elements	Models the frequency of elements in ranked data.

Hypergeometric(n_1, n_2, n_3)	Hypergeometric distribution	$f(x; n_1, n_2, n_3) = \frac{C(n_1, x) * C(n_2, n_3 - x)}{C(n_1 + n_2, n_3)}$	n_1 = population size; n_2 = number of successes in population; n_3 = sample size	Used for sampling without replacement.
Negative hypergeometric(n_1, n_2, n_3)	Negative hypergeometric distribution	$f(x; n_1, n_2, n_3) = \frac{C(n_1 + x - 1, x) * C(n_2 + n_3 - x - 1, n_3 - x)}{C(n_1 + n_2 + n_3 - 1, n_3)}$	n_1 = population size; n_2 = number of failures in population; n_3 = sample size	Used for modeling the number of trials until a specified number of successes.
Discrete Weibull(p, β)	Discrete Weibull distribution	$f(x; p, \beta) = p^{(x^\beta)} - p^{((x+1)^\beta)}$	p = parameter ($0 < p < 1$); β = shape parameter	Used for modeling discrete survival data.
Power series ($c, A(c)$)	Power series distribution	$f(x; c) = \frac{a_x * c^x}{A(c)}$	c = parameter; $A(c)$ = normalizing constant	A general class of discrete distributions.
Hypoexponential(a)	Hypoexponential distribution	$f(x; a) = \sum [p_i * \lambda_i * \exp(-\lambda_i x)]$	a = vector of parameters (p_i, λ_i)	Used in modeling systems with multiple exponential phases.

Hyperexponential(a)	Hyperexponential distribution	$f(x; a) = \sum [p_i * \lambda_i * \exp(-\lambda_i x)]$	a = vector of parameters (p_i, λ_i)	Used in modeling systems with multiple exponential phases.
Makeham(δ, κ, γ)	Makeham distribution	$f(x; \delta, \kappa, \gamma) = (\delta + \kappa e^{\gamma x}) e^{(-\delta x - (\kappa/\gamma) e^{\gamma x})}$	δ = base risk; κ = scale; γ = shape	Used in actuarial science and survival modeling.
Gompertz(δ, κ)	Gompertz distribution	$f(x; \delta, \kappa) = \delta \kappa e^{(-\kappa x)} e^{(-\delta e^{(-\kappa x)})}$	δ = shape parameter; κ = scale parameter	Used in modeling human mortality rates.
Kolmogorov-Smirnov(n)	Kolmogorov-Smirnov distribution	$f(x; n) = (1 - K_n(x))^n$	n = sample size	Used in goodness-of-fit tests for comparing distributions.

Noncentral $F(n_1, n_2, \delta)$	Noncentral F- distribution	$f(x; n_1, n_2, \delta) = \sum \left[\frac{e^{-\delta/2} (\delta/2)^k}{k!} \cdot \frac{n_1^{n_1/2 + k} n_2^{n_2/2} x^{n_1/2 + k - 1}}{B(n_1/2 + k, n_2/2) (n_2 + n_1 x)^{(n_1 + n_2)/2 + k}} \right]$	$n_1, n_2 =$ degrees of freedom; $\delta =$ noncentral ity parameter	Used in power analysis and noncentral hypothesis testing.
Doubly noncentral $F(n_1, n_2, \delta, \gamma)$	Doubly noncentral F- distribution	$f(x; n_1, n_2, \delta, \gamma) = \sum \left[\frac{e^{-(\delta + \gamma)/2} (\delta/2)^k (\gamma/2)^m}{k! m!} \cdot \frac{n_1^{n_1/2 + k} n_2^{n_2/2 + m} x^{n_1/2 + k - 1}}{B(n_1/2 + k, n_2/2 + m) (n_2 + n_1 x)^{(n_1 + n_2)/2 + k + m}} \right]$	$n_1, n_2 =$ degrees of freedom; $\delta, \gamma =$ noncentral ity parameters	Used in advanced power analysis and noncentral hypothesis testing.

$F(n_1, n_2)$	F-distribution	$f(x; n_1, n_2) = \frac{n_1^{n_1/2} n_2^{n_2/2} x^{(n_1/2 - 1)}}{(B(n_1/2, n_2/2) (n_2 + n_1 x)^{((n_1 + n_2)/2)})}$	$n_1, n_2 =$ degrees of freedom	Used in ANOVA and regression analysis.
Gamma-Poisson(α, β)	Gamma-Poisson distribution	$f(x; \alpha, \beta) = \frac{\Gamma(\alpha + x)}{\Gamma(x + 1) \Gamma(\alpha)} * (\beta / (1 + \beta))^x * (1 / (1 + \beta))^{\alpha}$	$\alpha =$ shape parameter; $\beta =$ rate parameter	Used for modeling overdispersed Poisson data.
Beta-binomial(a, b, n)	Beta-binomial distribution	$f(x; a, b, n) = C(n, x) * \frac{B(x + a, n - x + b)}{B(a, b)}$	$a, b =$ shape parameters ; $n =$ number of trials	Used for modeling overdispersed binomial data.
Beta-Pascal(n, a, b)	Beta-Pascal distribution	$f(x; n, a, b) = C(n + x - 1, x) * \frac{B(a + x, b + n)}{B(a, b)}$	$n =$ number of trials; $a, b =$ shape parameters	Used in Bayesian inference and modeling overdispersed count data.

Gamma-normal(μ, α, β)	Gamma-normal distribution	$f(x; \mu, \alpha, \beta) = \frac{\beta^\alpha}{\Gamma(\alpha) \sqrt{2\pi}} \cdot (x - \mu)^{\alpha-1} \cdot \exp(-\beta(x - \mu))$	μ = location parameter; α = shape parameter; β = rate parameter	Used in Bayesian inference and hierarchical modeling.
Logistic-exponential (a, β) S,V	Logistic-exponential distribution	$f(x; a, \beta) = \frac{\beta}{a} \cdot \frac{\exp(-\beta x)}{(1 + \exp(-\beta x))^2}$	a = scale parameter; β = shape parameter	Used in survival analysis and risk modeling.